

Module Accuracy Test Report

Postal Delivery Operations System - Bhandup West data set (26 beats, 130 deliveries)

Generated 2026-09-21 from measurements made by scripts/accuracy-test.ts. Nothing in this report is an assumed or typed-in result.

1. Summary

Module	What was measured	Result
Geocoding	Addresses the geocoder placed at street / building level	0 of 130 (0.0%)
Geocoding	Addresses that ended at area level, pincode level, or not found	79 area, 51 pincode, 0 not found
Beat assignment (production)	Deliveries put in the CORRECT beat / WRONG beat / left unassigned	0 (0.0%) / 5 (3.8%) / 125 (96.2%)
Beat assignment (alternative: locality name)	Locality resolves to exactly one beat and it is the right one / ambiguous	94 (72.3%) / 36 ambiguous
Route optimization (production ALNS pipeline)	Mean gap to the exact optimum on 20 small beat routes	0.00% (optimal on 100% of routes)
Route optimization (input order)	Same routes, deliveries visited in the order received	75.9% above optimum
Route optimization on larger rounds	Production gap to a long-search reference (25, 50, 100 stops)	3.2%, 2.0%, 3.1% (NN + 2-opt: 9.7%, 9.9%, 10.1%)

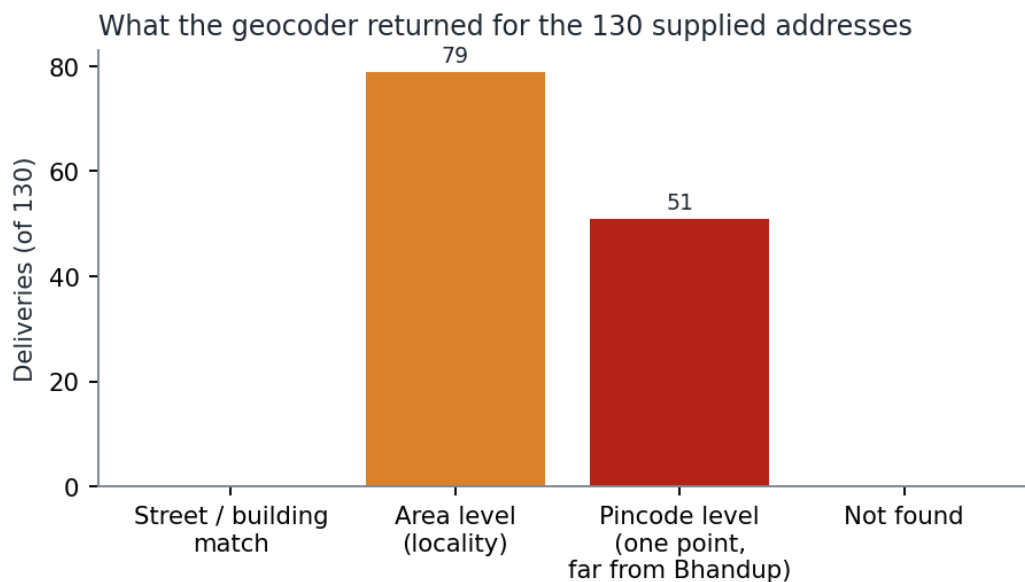
Reading this summary. The route optimizer is accurate: it finds the exact optimum on every small route tested and stays within a few per cent of a long-search reference on larger rounds, where every simpler algorithm is clearly further off. The weak link is the **address side**: the free OpenStreetMap geocoder placed **none** of the 130 chawl / building addresses at street level, and there is no official beat boundary, so automatic beat assignment from coordinates did not work on this data (0 correct). Matching the beat list's own locality names, by contrast, resolved 72.3% of the deliveries with no geocoding at all. Section 7 lists the limits of the current system.

2. Data used and how it was set up

- **Beat list** (26 beats, Set 1, Bhandup West SO): each row is Post Office, Set No., Beat No., Locality Name, Main Area Name, Pincode. It is a *name directory*: it says which localities and places belong to which beat but contains **no boundary and no coordinates**.
- **Delivery test file**: 130 deliveries, 5 per beat (name, phone, locality, main area, address). No coordinates, parcel weight or priority. The ground truth for every test is the beat the row belongs to in the beat list (delivery n belongs to beat n/5, rounded up).
- **System setup** (through the system's own beat import, verification, postman and delivery-import functions): all previous data was removed after a database backup; 26 beats were created, one postman with a login per beat, and the 130 deliveries were imported with the real geocoder and PostGIS matching.
- **Territories**: because the beat list has no boundaries, each beat's territory was inferred as a 350 m buffer around real OpenStreetMap locations found for its localities (20 of 26 beats had at least one; the rest need a territory drawn by an administrator). These are proposals, not surveyed boundaries.
- **Note on the beat list file**: the PDF was supplied in chat and was not available as a file, so beat-directory.csv holds every locality the PDF names for every beat, but only the main-area rows that also appear in the delivery file. Nothing was invented.

3. Test 1 - Geocoding accuracy

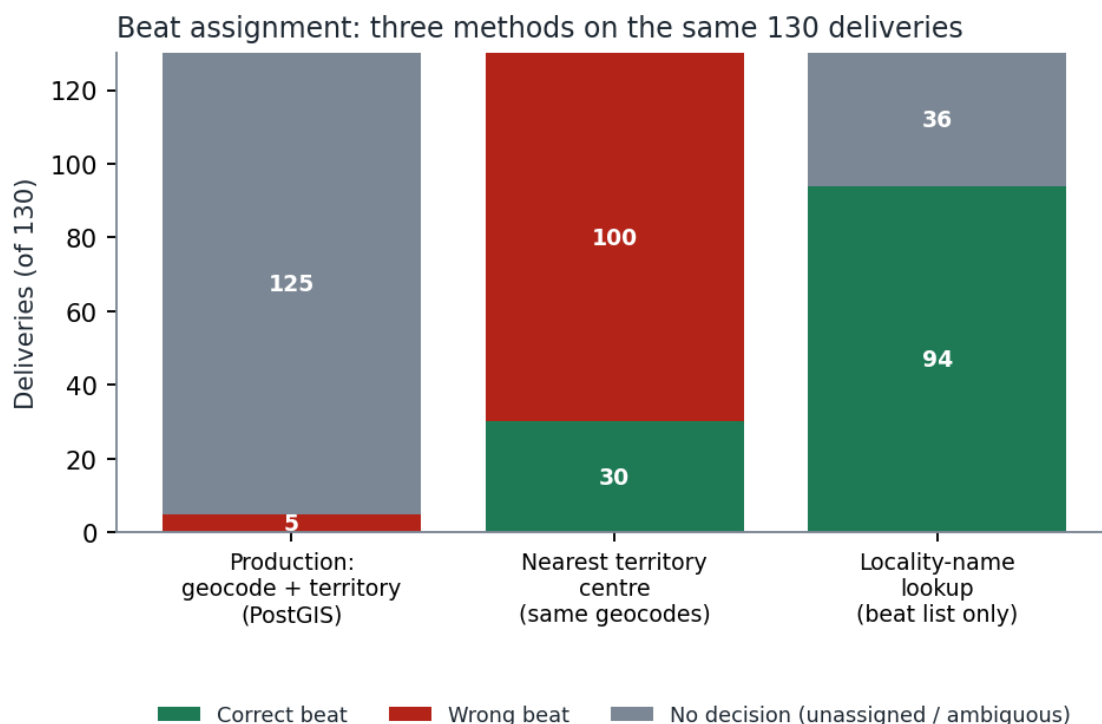
The system geocodes an address with OpenStreetMap Nominatim, trying the full address first, then just the area, then just the pincode (each fallback is tagged with a lower confidence).



Of 130 addresses, 0 (0.0%) were placed at street/building level, 79 at area level, and 51 only at pincode level; 0 were not found. Only 17 distinct coordinates were produced for the 130 addresses that got a position, so many deliveries share one point: all 51 pincode-level results are the same single point, about 12 km from the Bhandup West post office. Only 50 of the 130 positions fall inside the Bhandup West area.

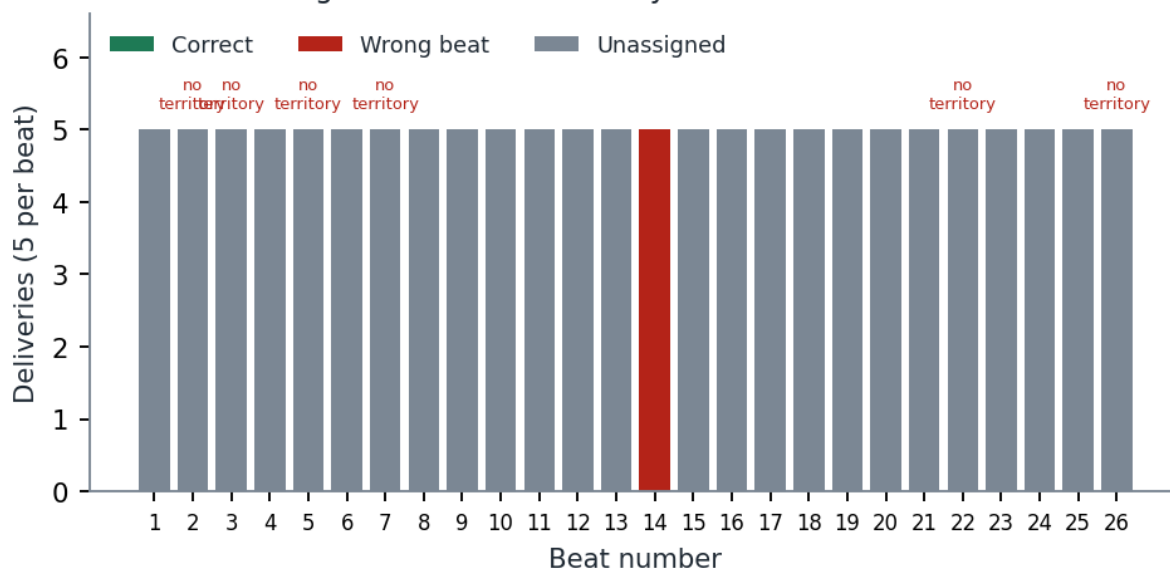
4. Test 2 - Beat assignment accuracy (three methods)

Production method: geocode, then find the verified beat territory that contains the point (PostGIS). **Alternative A:** pick the beat whose territory centre is nearest to the geocoded point. **Alternative B:** look the locality name up in the beat list (needs no geocoding). A and B are evaluated offline on the same 130 deliveries; they are not in production.



Production: 0 correct, 5 wrong, 125 left unassigned (exceptions raised: no beat match 80, multiple beat match 45). Nearest centre: 30 correct, 100 wrong. Locality lookup: 94 resolved correctly, 36 ambiguous (the locality belongs to several beats), 0 not in the list.

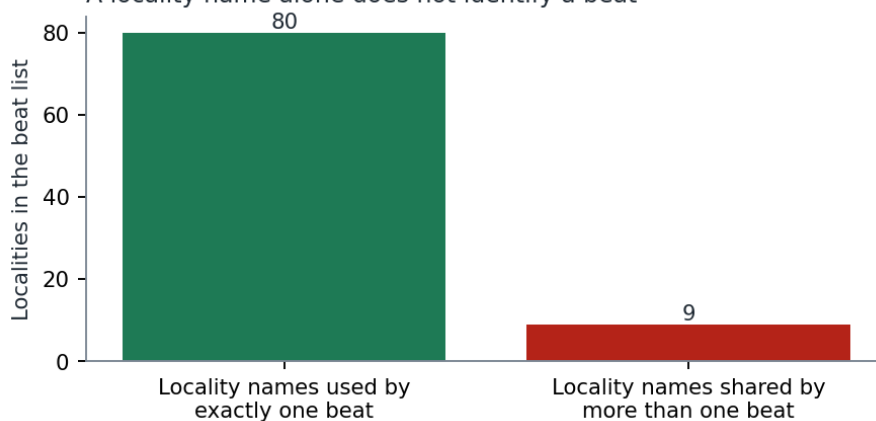
Production assignment result for every beat



Geocode quality	Correct	Wrong	Unassigned
area level	0	5	74
pincode level	0	0	51

Territories: 20 of 26 beats have one; 68 pairs of territories overlap, so a point can fall in more than one beat (the system then refuses to guess and raises an exception).

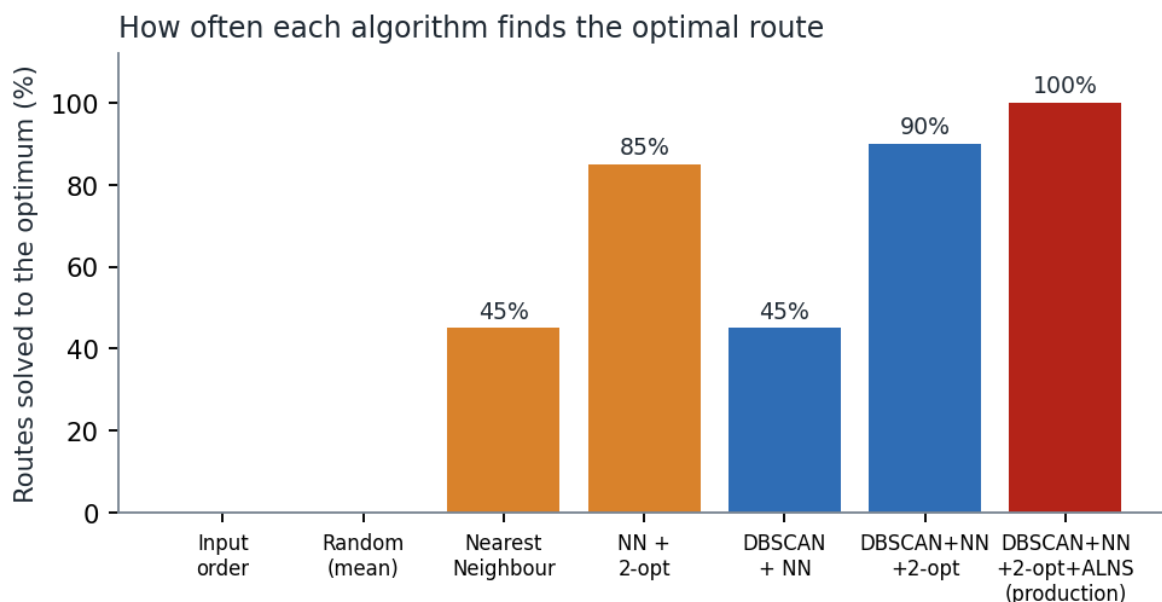
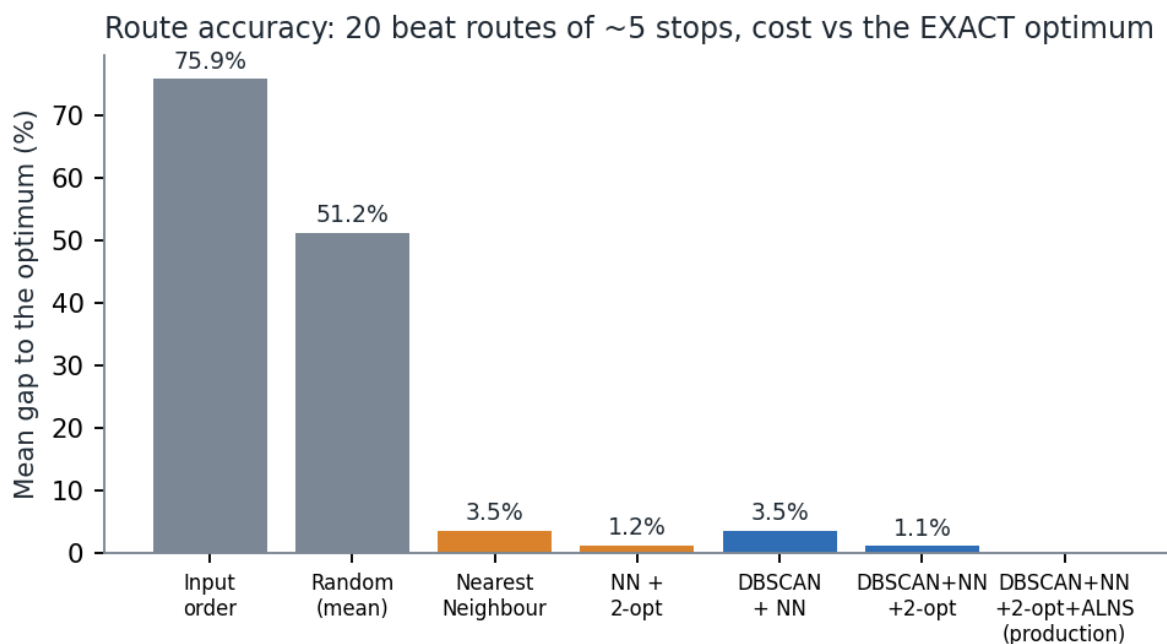
A locality name alone does not identify a beat



5. Test 3 - Route optimization accuracy (algorithm comparison)

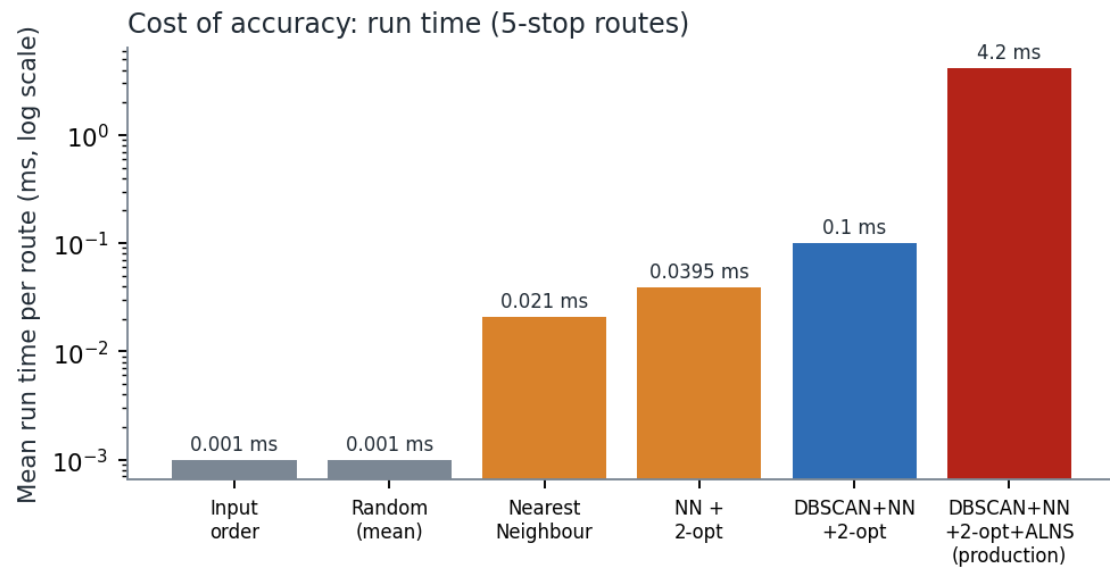
Every algorithm plans the same delivery set on the same **real road travel-time matrix (OSRM)** and is scored by the same production cost function (driving time + load penalty + priority penalty, service time included). **Accuracy = gap to the optimum**: for routes of up to 10 stops the exact optimum was found by exhaustive search, so 0% means the best possible route. For larger rounds there is no exact answer; the reference is the cheapest route found by any of the algorithms or by five independent long ALNS runs (15 000 iterations, different seeds, clusters not forced together), so the production pipeline is not compared with itself and can show a gap above 0%.

Where the points come from. The supplied addresses cannot be geocoded to buildings (Test 1), so route tests use points *placed* deterministically within 300 m of the OpenStreetMap anchor of the delivery's beat (20 beats had an anchor). The road network, distances and travel times are real; the exact house positions are simulated. The comparison between algorithms is valid; the absolute minutes are illustrative.

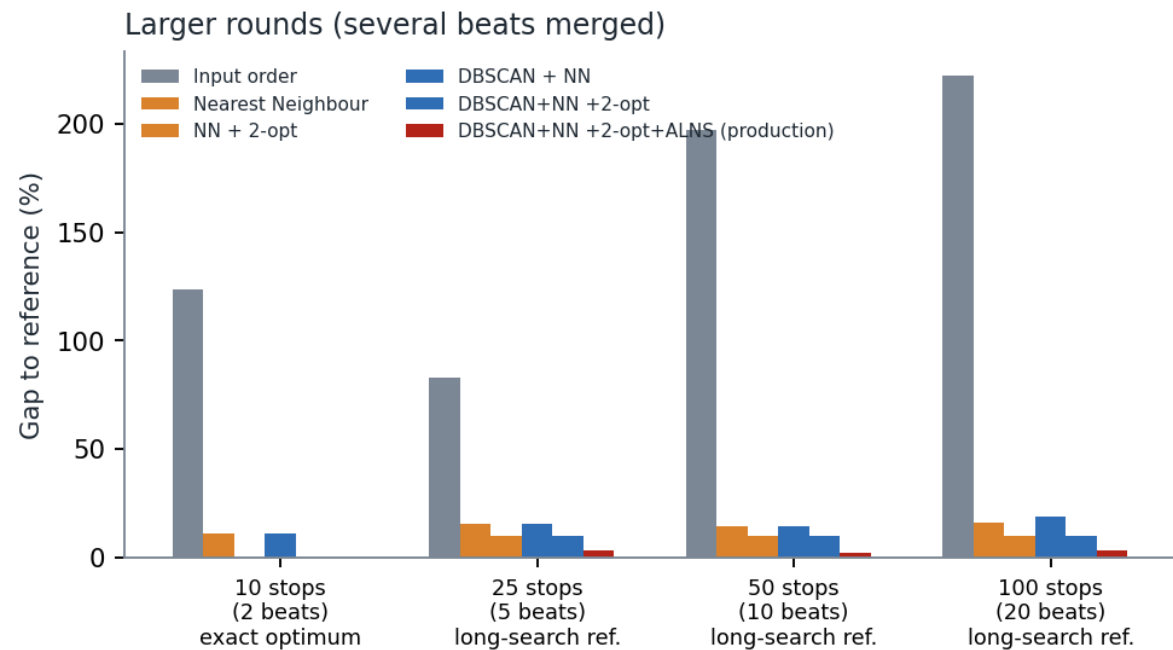


Algorithm	Mean gap	Worst gap	Optimal	Mean km	Mean min	Run time
Input order	75.87%	123.97%	0%	11.36	22.7	0 ms
Random order (mean of 30)	51.24%	80.02%	0%	-	-	0.0005 ms
Nearest Neighbour	3.54%	19.48%	45%	6.19	13.2	0.021 ms
NN + 2-opt	1.16%	19.23%	85%	6.01	12.9	0.0395 ms

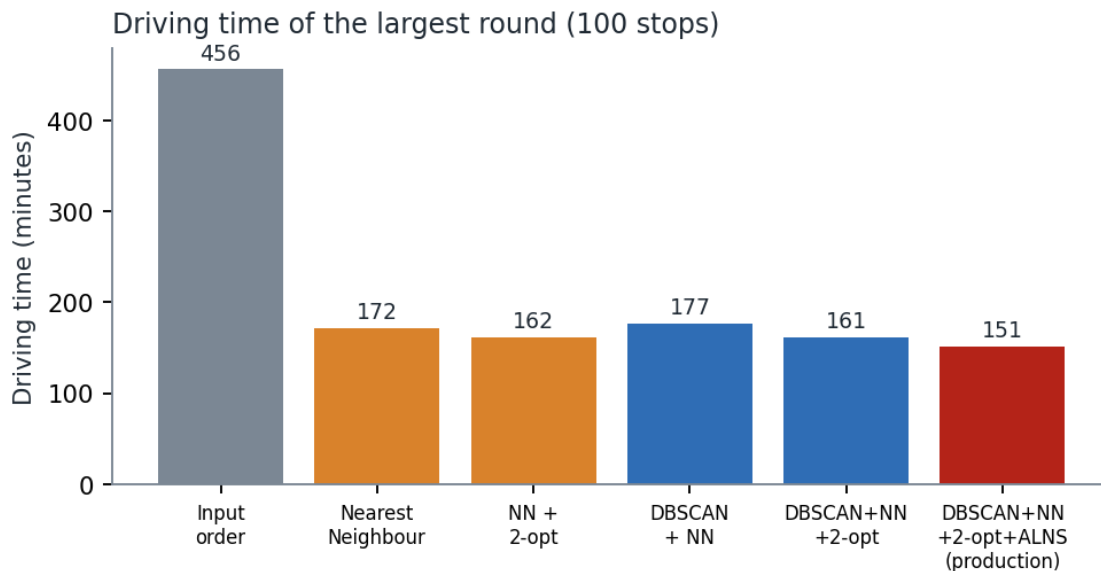
Algorithm	Mean gap	Worst gap	Optimal	Mean km	Mean min	Run time
DBSCAN + NN	3.54%	19.48%	45%	6.19	13.2	(in next row)
DBSCAN + NN + 2-opt	1.12%	19.23%	90%	5.98	12.9	0.1 ms
DBSCAN + NN + 2-opt + ALNS (production)	0.00%	0.00%	100%	5.88	12.8	4.2 ms



Larger rounds



Round	Reference	Input order	Nearest Neighbour	NN + 2-opt	DBSCAN + NN	DBSCAN+NN +2-opt	DBSCAN+NN +2-opt+ALNS (production)
10 stops (2 beats)	exact	123.47%	10.89%	0.00%	10.89%	0.00%	0.00%
25 stops (5 beats)	long search	82.94%	15.38%	9.69%	15.38%	9.69%	3.23%
50 stops (10 beats)	long search	197.19%	14.39%	9.87%	14.39%	9.87%	1.99%
100 stops (20 beats)	long search	222.24%	16.24%	10.08%	19.12%	9.69%	3.11%



On the largest round (100 stops) the production pipeline drives 151 minutes / 57.0 km against 456 minutes / 219.6 km for the input order. On the 5-stop beat routes, plain Nearest Neighbour + 2-opt already finds the optimum on 85% of routes, so those small routes barely separate the good algorithms; the differences appear from 25 stops upward.

6. Problems found in the supplied data

- 9 of the 89 locality names in the beat list appear under more than one beat (for example MANGATRAM PETROL PUMP TO GANDHI NAGAR: beats 1, 23; VILLAGE ROAD: beats 2, 5; SUBHASH NAGAR: beats 4, 24). A delivery's locality alone therefore cannot decide its beat; 36 of the 130 test deliveries sit in such a locality.
- Beat 12 (BHATTIPADA, KESHAVJI NAGAR, JANGAL MANGAL ROAD) is listed with pincode 40078 (5 digits) in the supplied beat list.
- Truncated / incomplete main-area text in the delivery file: "WING 23 FLOORS".
- 110 recipient names repeat across the 130 deliveries (only 20 distinct names), so a recipient name cannot identify a delivery.
- The delivery file has no parcel count, weight or priority, so the load and priority parts of the route cost are not exercised by this data.
- The beat list has no boundaries or coordinates; every territory is an inference.

7. Limitations of the current system

Address and beat matching

- **Geocoding coverage.** The free OpenStreetMap geocoder does not know most chawl, building, shop-line or 'GALA' names: only 0.0% of these addresses were placed at street/building level. The rest fall back to area or pincode level, so many deliveries share one point.
- **Geocode confidence is not checked when picking a beat.** The 51 pincode-level results share one point ~12 km away and were left unassigned only because no territory contains that point. Area-level points (a whole locality reduced to one point) were accepted like exact ones: 5 such deliveries went to the wrong beat, and the rest fell in gaps or in overlaps between territories.
- **No official beat boundaries.** Territories are inferred from a few known place names and overlap (68 overlapping pairs); 6 of 26 beats have no territory at all and cannot receive deliveries automatically.
- **Assignment is purely spatial.** The system does not use the beat list's own locality / main-area names to assign a delivery, even though that directory is the authoritative source; a name-based match could resolve 72.3% of these deliveries unambiguously with no geocoding.
- **The beat list format is not imported as-is.** The beat import expects one row per beat with a territory; the supplied format has hundreds of rows per beat and no territory, and a PDF cannot be uploaded (CSV / XLSX only). The set-up scripts in this project convert it.

- **Public geocoder limits.** Nominatim allows about one request per second and forbids bulk use: importing 130 addresses took several minutes; a daily post-office volume needs an own geocoder or a paid service.

Route optimization

- **Travel times are typical, not live.** OSRM road times contain no live traffic or time-of-day effects; the public OSRM demo server is used and is not meant for production load.
- **The optimizer is a heuristic.** DBSCAN, Nearest Neighbour, 2-opt and ALNS give no guarantee of optimality: on the larger rounds the production route was 3.2%, 2.0%, 3.1% above the best route a longer search found.
- **Clusters are kept whole by default.** Keeping each DBSCAN cluster as one run keeps neighbourhoods together but costs quality: on the 100-stop round plain NN + 2-opt (no clustering) is 10.1% from the reference while DBSCAN + NN + 2-opt is 9.7%. It is a business choice, switchable by `ROUTE_ALNS_PRESERVE_CLUSTERS`.
- **No time windows, vehicle capacity, breaks or multiple postmen.** One postman, one round, one start point; there is no balancing of work between beats or postmen.
- **Priority and parcel weight are optional inputs** that this data set did not supply, so their effect on routes is untested here.
- **Route tests used placed points** (Section 5): the ranking of algorithms is reliable, the absolute minutes are not a forecast for these real addresses.

Data and operations

- Ground truth for the beat of each delivery is taken from the file layout (5 deliveries per beat); there are no independent coordinates to measure geocoding distance error.
- Only one post office and one 130-delivery batch were tested; scaling to a full office day has not been measured beyond 100-130 stops per round.
- Accuracy of the postman app's field behaviour (GPS, offline queue) is outside this report.

8. Recommendations

- Add a **directory-based matcher** that uses the beat list's locality + main-area names first (with fuzzy matching), and geocoding only as a fallback.
- Do not auto-assign on **pincode-level or area-level** geocodes: send them to the exceptions list, or accept an administrator-drawn beat.
- Give every beat a **surveyed territory** (draw the 26 in the Beats map) and verify them; overlapping boundaries should be resolved by the administrator.
- Use a **dedicated geocoder** (self-hosted Nominatim/Pelias or a paid API) and collect house coordinates from postmen once, then reuse them.
- Run OSRM on the office's own server, and add priority / parcel data to the imports so the full cost model is used.

9. Reproducing the tests

```
python backend/data/bhandup/build_data.py
npx tsx scripts/setup-bhandup.ts wipe | beats | deliveries
npx tsx scripts/accuracy-test.ts
python scripts/make_accuracy_report.py
```

Postman logins created: beat01.postman@postal.local ... beat26.postman@postal.local, password ChangeMe123! (administrator: admin.bhandup@postal.local).